

	$\frac{NA2\pi r}{120 \times 1.3 \times 10^{-2} \times 2\pi \times 25}$															
5(a)	Gravitational potential at a point in a gravitational field is the work done per unit mass by and external agent in bringing a small point mass from infinity to that point without a change in KE.															
5(b)(i)	$\phi = -\frac{GM}{r}$$M = \frac{-\phi r}{G}$<table><tr><th>Dist from centre of Jupiter</th><th>Grav Potential</th><th>Mass of Jupiter/10^{27}kg</th></tr><tr><td>2.0</td><td>-6.4</td><td>1.92</td></tr><tr><td>4.0</td><td>-3.2</td><td>1.92</td></tr><tr><td>6.0</td><td>-2.1</td><td>1.89</td></tr><tr><td>8.0</td><td>-1.6</td><td>1.91</td></tr></table>	Dist from centre of Jupiter	Grav Potential	Mass of Jupiter/ 10^{27} kg	2.0	-6.4	1.92	4.0	-3.2	1.92	6.0	-2.1	1.89	8.0	-1.6	1.91
Dist from centre of Jupiter	Grav Potential	Mass of Jupiter/ 10^{27} kg														
2.0	-6.4	1.92														
4.0	-3.2	1.92														
6.0	-2.1	1.89														
8.0	-1.6	1.91														

5(b)(ii)	Gravitational potential energy = $\frac{-GMm}{r} = \frac{-(6.67 \times 10^{-11})(1.91 \times 10^{27})(8.93 \times 10^{22})}{4.22 \times 10^8}$ = $-2.6959 \times 10^{31} \text{ J}$ Kinetic energy = $\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{2\pi r}{T}\right)^2$ = $\frac{1}{2}(8.93 \times 10^{22})(2\pi \times 4.22 \times 10^8 / 1.53 \times 10^5)^2$ = $1.3410 \times 10^{31} \text{ J}$ Total energy = $E_k + E_p = -1.35 \times 10^{31} \text{ J}$
5(c)	$1.0 \times 10^{-10} \text{ m}$ from S is so far relative to scale of graph that the electric potential V is effectively zero. At a distance of $1.4 \times 10^{-15} \text{ m}$ from S electric potential is $1.02 \times 10^6 \text{ J C}^{-1}$. Loss in proton's KE is equal to gain in its electric potential energy. $\frac{1}{2}mv^2 - 0 = eV$ $v = \sqrt{\frac{2eV}{m}}$ = $\sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 1.02 \times 10^6}{1.67 \times 10^{-27}}}$ = $1.4 \times 10^7 \text{ ms}^{-1}$
5(d)	Similarity: The magnitude of the potential is inversely proportional to distance from centre of respective object. Difference: Gravitational potential is always negative whereas electric potential can be positive or negative depending on the polarity of the charge.
6(a)	The rate of change of A decreases with t since the gradient decreases with t